CHE210: *DISCUSSION SESSION 10* **NAME:**___________________________________

NOVEMBER 5, 2019

Approximation of Standard-State $\Delta\hat{H}_r^{\circ}$ *[Murphy-P6.36]*

We need to calculate a standard-state $\Delta\hat{H}_r^{\circ}$ for the oxidation of *p*-xylene (C_8H_{10}) to terephthalic acid (TPA, $C_8O_4H_6$), which is used in making plastic 2-L soda bottles. However, heats of formation of the compounds from the respective elements are not available. References tables do have information on heats for combustion to CO_2 (g) and H_2O (l) at 25°C, and show that the energy released is 1089 kcal/mol for *p*-xylene (l) and 770 kcal/mol for TPA (s).

$$C_8H_{10}(l) + 3\ O_2(g) \rightarrow\ C_8O_4H_6(s) + 2\ H_2O(l)$$

Determine a numerical value for $\Delta\hat{H}_r^{\circ}$. Given that $\Delta\hat{H}_f^{\circ}$ for CO_2 = -94 kcal/mol and $\Delta\hat{H}_f^{\circ}$ for liquid H_2O = -68 kcal/mol, calculate $\Delta\hat{H}_f^{\circ}$ for *p*-xylene and TPA.